Sharini Rao

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Data Scientist & ML Engineer — Bio-Inspired Systems and Design

Education

University of California, Berkeley Berkeley, CA

B.A. Data Science (Domain Emphasis: Cognition) — Berkeley Design Innovation Certificate *Expected May 2027*

- **Technical Core:** Data Engineering (Data 101), Data Mining & Analytics (Data 144), Probability & Statistics (Data 140), UI Design & Development (CS 160), Data Structures (CS 61B), Data & Computational Structures (Data C88C).
- **Bio-Systems & Cognitive Health Depth:** Bio-Inspired Design (INTEGBI 130), Health Economics (PB HLTH 126), Game Theory (Econ 110), Organic Chemistry, Chemical Biology, Quantitative Public Health.

Technical & Domain Skills

Biological & Systems Analytics: Spectrophotometry, Spectroscopic Modeling, Kinetic Modeling (Pseudo-1st/2nd Order), Biomaterial Characterization, Experimental Design (DoE), Real-Time Sensor Telemetry
Languages & Data: Python, SQL (Postgres/MySQL), R, Java, TypeScript, JavaScript, HTML/CSS, MATLAB
Frameworks & Pipelines: Next.js, React, Node.js, Redis, Pusher, REST APIs, RAG, Sentence-Transformers, Pandas, Git, Vercel, WCAG Accessibility, Figma

Experience

UC Berkeley Imaging Systems Laboratory Berkeley, CA

Undergraduate Researcher — Ti Lab & Conolly Lab *May 2024 – Aug 2026*

- Functionalized magnetic nanoparticles with DNA aptamer biosensors, modeling metabolite binding dynamics (potassium, dopamine) for diagnostic data pipelines (Ti Lab).
- Synthesized superparamagnetic iron oxide nanoparticles to improve MRI magnetic resolution 10-fold; processed experimental biosensor datasets and presented findings on DNA-origami nanoparticle design (Conolly Lab).

Legali AI Berkeley, CA

Software Engineering Intern *May 2026 – Present*

- Shipped 7 production web applications using Next.js, React, Node.js, and Figma-to-code workflows, merging 6 PRs in Git while maintaining production-grade platform reliability.
- Engineered a 12-question interactive workflow end-to-end featuring multi-step state management, Web Share API integration, and WCAG-compliant accessibility across 7 languages.

UC Berkeley Haas School of Business Berkeley, CA

Data Analytics Research Apprentice *Jan 2026 – Present*

- Built automated ETL pipelines in Python (Pandas) to clean survey datasets for NGO health equity research, cutting manual preprocessing lead time by 4 weeks.
- Analyzed survey responses from 1,200+ participants using EDA, producing data visualizations and stakeholder-ready insights that shaped health research priorities.

Featured Projects

NutriPath: RAG-Grounded Clinical Meal Planning Assistant | *AI & Health Systems* Personal Project

- Built a retrieval-augmented generation pipeline over clinical rules using sentence-transformer embeddings and cosine search backed by Redis, achieving 72% top-1 and 90% top-3 retrieval accuracy.

Lignocellulosic Bio-Adsorbancy & Circular Upcycling | *Materials Analytics & Green Chem* Dec 2025

- Evaluated whole vs. crushed pistachio shell lignocellulosic matrix for organic pollutant (Crystal Violet) removal across pH conditions, demonstrating 84.34% removal efficiency via pseudo-second-order kinetics ($R^2 = 0.999$).

FedUp: Real-Time Algorithmic Group Dining & Recommendation System | *Full-Stack / Systems* Personal Project

- Built a full-stack Next.js/Postgres application deploying a preference recommendation engine via Google Places and Claude APIs, integrating Redis/Pusher for real-time synchronized voting.

Leadership & Community

Society of Women Engineers (SWE) Berkeley, CA

Evening with Industry Committee *Aug 2025 – Present*

- Organized Berkeley's largest SWE networking event, coordinating outreach and logistics for 15–20 corporate partner companies, keynote speakers, and 100+ student attendees.

Kids Cooking for Life Oakland, CA

Volunteer Culinary & Nutrition Educator *Jan 2024 – May 2024*

- Taught nutrition principles and culinary techniques across 9 recipes to classes of 15–20 students at KIPP Bridge School, translating dietary science into accessible community health education.

UC Berkeley Rec Well Berkeley, CA

FitStaff Operations Lead *May 2024 – Present*

- Coordinate daily operations and team workflows for a high-traffic campus facility, managing shift scheduling and real-time operational problem solving for 2.5+ years.